

2nd order linear ODE's

$$a(x) y'' + b(x) y' + c(x) y = d(x)$$

standard form

$$y'' + p(x) y' + q(x) y = r(x)$$

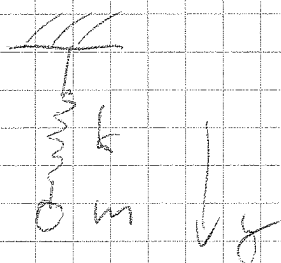
Homogeneous case

$$r(x) = 0$$

$$y'' + p(x) y' + q(x) y = 0 \quad (*)$$

- Example

mass and spring



$$m y'' = -k y$$

$$\text{let } m=1, k=1$$

$$y'' + y = 0$$

• What are the solutions?

$$\sin(x), \cos(x) \quad \text{or} \quad e^{ix}, e^{-ix}$$

• How many initial conditions do we need?

need both $y(t_0)$ and $y'(t_0)$

$$\text{let } y(0) = 1$$

$$y'(0) = 1$$

$$y(x) = A \cos x + B \sin x$$

$$y'(x) = -A \sin x + B \cos x$$

$$y(0) = A$$

$$y'(0) = B$$

$$\Rightarrow A = B = 1$$

- Superposition principle

If $y_1(x)$ and $y_2(x)$ are both solutions of $(*)$, then so is any linear combination $\alpha y_1(x) + \beta y_2(x)$

- Linear independence

Def.: Two functions are linearly dependent if $\exists k_1, k_2 \neq 0$ such that

$$k_1 y_1(x) + k_2 y_2(x) = 0 \quad (\forall x)$$

If no such k_1 and k_2 exist, they are linearly independent.

Remark: This generalizes to more than two functions.

For 2 functions, equivalent to $\exists \alpha$ s.t. $y_2(x) = \alpha y_1(x)$

- Wronskian

Let y_1 and y_2 be solutions of (*)

y_1 and y_2 are linearly dependent
 $\iff$ c.f.f

$$\exists x \text{ s.t. } W(x) = y_1(x)y_2'(x) - y_1'(x)y_2(x) = 0$$

Remark: it is a determinant $W(x) = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix}$

Proof $\Downarrow$ direction:

linearly dependent $y_2(x) = \lambda y_1(x)$

$$W(x) = y_1(x)\lambda y_1'(x) - y_1'(x)\lambda y_1(x) = 0$$

other direction proof optional reading in the book.

- Initial value problem and general solution

The general solution of

$$y'' + p(x)y' + q(x)y = r(x)$$

is given by

$$y(x) = c_1 y_1(x) + c_2 y_2(x) + y_p(x)$$

where y_p is any particular solution, and

y_1 and y_2 are linearly independent solutions of the homogeneous eqn. (*).

The IVP $y(x_0) = A$
 $y'(x_0) = B$

can be solved for c_1 and c_2 .

Reduction of order

Assume that we know one solution y_1 of

$$y'' + p(x)y' + q(x)y = 0$$

How to find the other?

Assume $y_2(x) = u(x)y_1(x)$

$$y_2' = u'y_1 + uy_1'$$

$$y_2'' = u''y_1 + 2u'y_1' + uy_1''$$

$$u''y_1 + 2u'y_1' + uy_1'' + p(x)u'y_1 + p(x)uy_1' + q(x)uy_1 = 0$$

$$u''y_1 + u'(2y_1' + py_1) + u(\underbrace{y_1'' + py_1' + q}_{=0}) = 0$$

y_1 is solution

$$u'' + u' \left(2 \frac{y_1'}{y_1} + p \right) = 0$$

Let $u' = u$

$$u' + u \left(2 \frac{y_1'}{y_1} + p \right) = 0$$

1st order separable ODE

$$\int \frac{1}{u} du = - \int \left(2 \frac{y_1'}{y_1} + p \right) dx$$

$$\ln u = -2 \ln y_1 - \int p dx$$

$$u = \frac{1}{y_1^2} e^{-\int p dx}$$

$$u = \int u dx$$

Homogeneous linear 2nd order ODE with constant coefficients

$$y'' + ay' + by = 0$$

Ansatz: $y(x) = e^{\lambda x}$

$$\lambda^2 e^{\lambda x} + a \lambda e^{\lambda x} + b e^{\lambda x} = 0$$

$$(\lambda^2 + a\lambda + b) e^{\lambda x} = 0$$

$$\lambda^2 + a\lambda + b = 0$$

$$\lambda_{1/2} = \frac{-a \pm \sqrt{a^2 - 4b}}{2}$$

$$y_1(x) = e^{\lambda_1 x}$$

$$y_2(x) = e^{\lambda_2 x}$$

linearly independent if $\lambda_1 \neq \lambda_2$

1. Case $\lambda_1 \neq \lambda_2$, real

$$y(x) = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

sum of two exponentials

3. Case $\lambda_1 \neq \lambda_2$ complex

$$\lambda = -\frac{a}{2} \pm i\omega \quad \text{with} \quad \omega = \sqrt{b - \frac{a^2}{4}}$$

$$y(x) = e^{-\frac{ax}{2}} (A \cos \omega x + B \sin \omega x)$$

2. case $\lambda_1 = \lambda_2 = -\frac{a}{2}$ double root

$$y_1 = e^{-\frac{ax}{2}}$$

we need a linearly independent solution
look as

$$y_2 = u y_1$$

...

$$u'' y_1 + u' \underbrace{(2y_1' + a y_1)}_0 + u \underbrace{(y_1'' + a y_1' + b y_1)}_0 = 0$$

y_1 is solution

$$2y_1' = -a e^{-\frac{ax}{2}} = -a y_1$$

$$u'' y_1 = 0$$

$$u'' = 0$$

$$u = \underbrace{c_1 x + c_2}$$

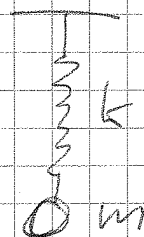
choose this part so y_2 is lin. indep.

$$y_2 = x e^{-\frac{ax}{2}}$$

$$\underline{y(x) = (c_1 + c_2 x) e^{-\frac{ax}{2}}}$$

Example Damped harmonic oscillator

$$m \ddot{y} = -ky - \underset{\substack{\uparrow \\ \text{damping}}}{c \dot{y}}$$



$$\ddot{y} + \frac{c}{m} \dot{y} + \frac{k}{m} y = 0$$

$$y(t) = e^{\lambda t}$$

$$\lambda^2 + \frac{c}{m} \lambda + \frac{k}{m} = 0$$

$$\lambda = \frac{1}{2} \left(-\frac{c}{m} \pm \sqrt{\frac{c^2}{m^2} - 4 \frac{k}{m}} \right)$$

Case 1 overdamped

$$\frac{c^2}{m^2} > 4 \frac{k}{m}, \lambda \in \mathbb{R}$$

$$y(t) = A e^{-\lambda_1 t} + B e^{-\lambda_2 t}$$

Case 2 critical

$$y(t) = (c_1 + c_2 x) e^{-\frac{c}{2m} t}$$

Case 3 underdamped

$$y(t) = e^{-\frac{c}{2m} t} (A \cos \omega t + B \sin \omega t)$$

$$\omega^2 = \frac{k}{m} - \frac{c^2}{4m^2}$$

$$\omega_0 = \sqrt{\frac{k}{m}}$$

